Python Data Types

1. Numeric Data Types

a) int (Integer)

Whole numbers (positive, negative, or zero)

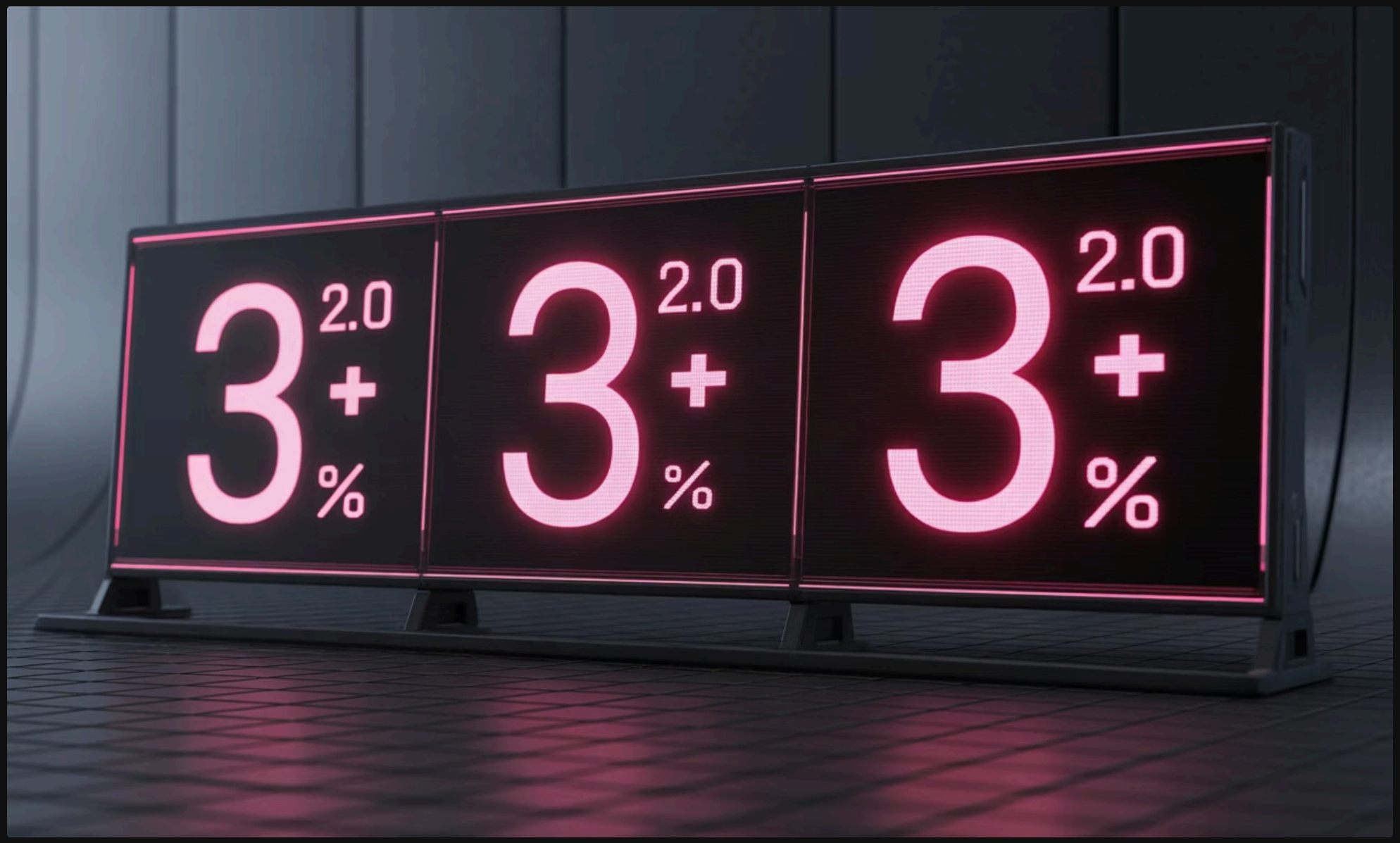
```
x = 25
print(type(x))  # <class 'int'>
```

b) float (Decimal Number)

```
y = 3.1415
print(type(y))  # <class 'float'>
```

c) complex (Numbers with j)

```
z = 2 + 5j
print(type(z))  # <class 'complex'>
```



2. String (str)

Used for text

Written inside `' '` or `" "`

```
name = "Python"
print(type(name))  # <class 'str'>
```

String is index-based

```
name = "Python"
print(name[0])   # P
print(name[-1])  # n
```

3. Boolean (bool)

```
a = True
b = False
print(type(a))  # <class 'bool'>
```

Used mostly in conditions:

```
print(10 > 5)  # True
```

```
print(10 < 5)  # False
```

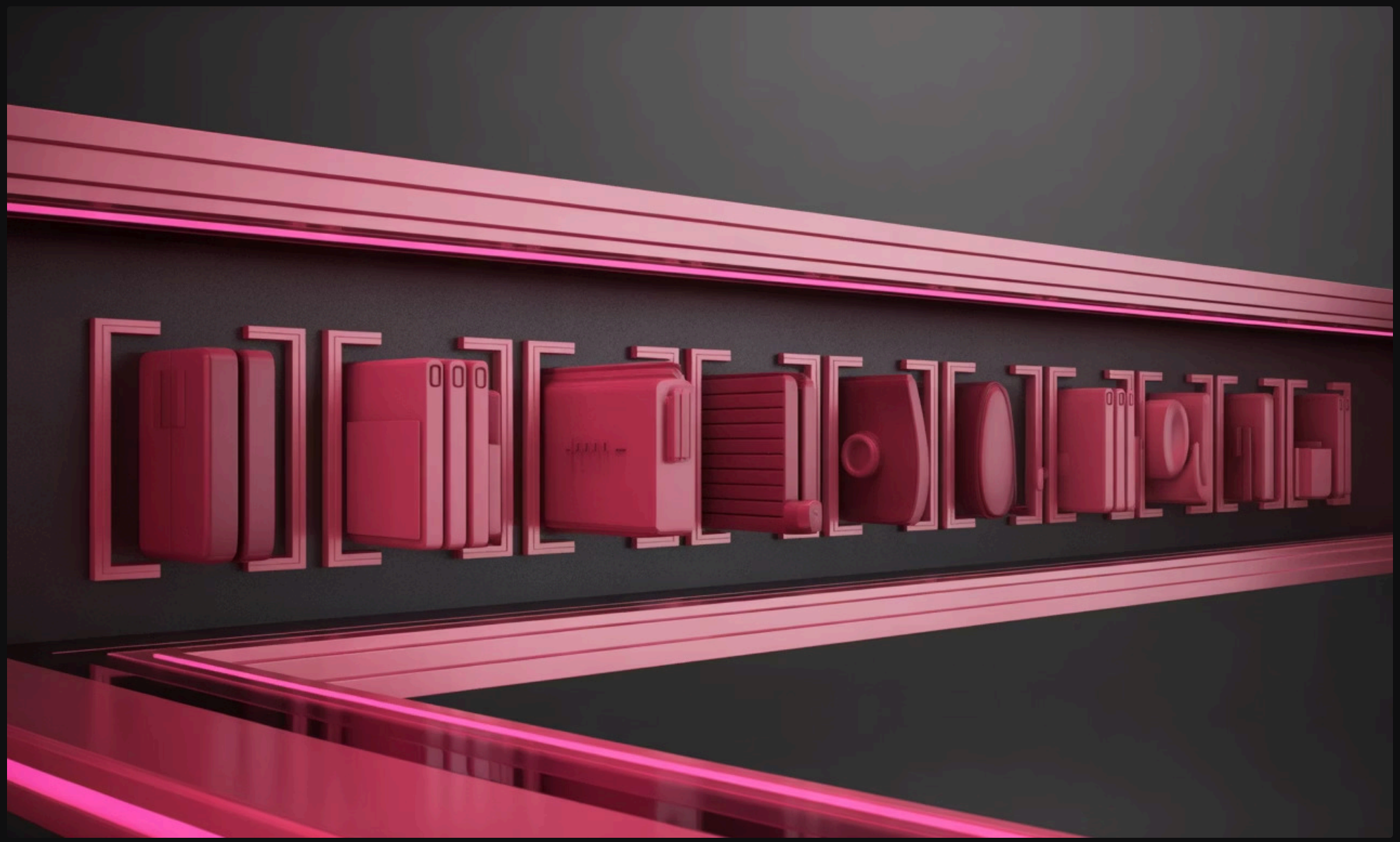
4. List

Ordered

Changeable

Allows duplicates

```
fruits = ["apple", "banana", "cherry"]
fruits[1] = "mango"  # changing value
print(fruits)
```



5. Tuple

- Ordered
- **NOT changeable (Immutable)**

```
colors = ("red", "green", "blue")

# colors[0] = "yellow" #
```

6. Set

Unordered

No duplicate items

Items cannot be indexed by position

```
a = {1, 2, 3, 3, 2}
print(a)  # {1, 2, 3}
```

7. Dictionary (dict) & 8. NoneType

Dictionary (dict)

Key–Value pairs

```
student = {
    "name": "Naveen",
    "age": 21,
    "marks": 85
}
print(student["name"])  # Naveen
```

NoneType

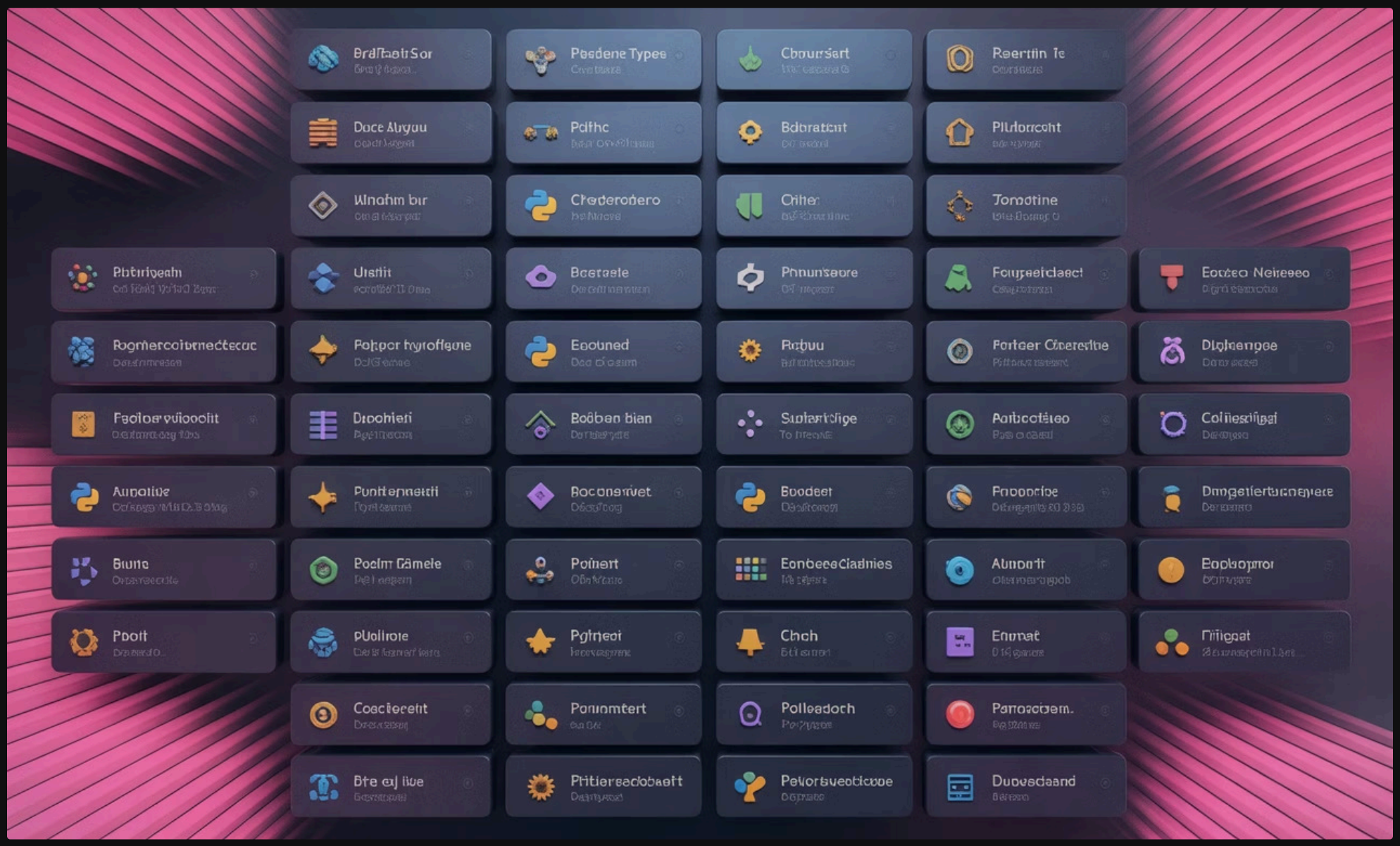
Represents Empty / No value

```
x = None
print(type(x))  # <class 'NoneType'>
```

⭐ Data Type Conversion (Type Casting)

```
a = "10"
b = int(a)  # string → int
print(b + 5)  # 15
```

```
x = 5
y = float(x) # int → float
print(y) # 5.0
```

🎯 Quick Revision Table

Type	Example	Mutable?	Ordered?
int, float, bool, str	10, 2.5, True, "Hi"	❌	✅
list	[1, 2, 3]	✅	✅
tuple	(1, 2, 3)	❌	✅
set	{1, 2, 3}	✅	❌
dict	{"name": "Amit"}	✅	✅